

FINAL TERM MCQS (BT401)

PREPARE BY: ASHAM

TOPIC NO# 2

1. PGRFA stands for.....**plant genetic resource for food and agriculture**
2. Animal genetic resource are (Animal genetic resource, Animal genetic resource for food and agriculture, 3.farm animal genetic resource) **all of these**

TOPIC NO# 3

3. Genetic resources are sometimes called_ **first resource**
4.are link from generation to generation of all living matter.....**genes**

TOPIC NO# 4-5

5. Landraces is a__ **local variety of domesticated plants**
6. ____are varieties developed since the advent of scientific agriculture in the late nineteenth century and that are no longer cultivated..... **Obsolete varieties**
7. The sum total of all the gene present in a crop and its related species constitutes its.....**germplasm**

TOPIC NO#6

8. Conservation of plant genetic resource is necessary for..... **Food scrutiny and Agro-biodiversity (both A&B)**

TOPIC NO#7

9. Special care unit..... **ex situ**
10. ex-situ conservation include **all given (Seed Gene bank, Botanical gardens, zoological parks)**
11. _____Is an example of an ex-situ conservation.....**seed bank**
12. Gene banks are also known as..... **Germplasm banks.**

LAST FIGHT OF WARRIORS

13. Field gene banks also called plant gene banks are areas of land in which _____collections of growing plants are assembled **germplasm**
14. Gene bank is a type of **ex-situ conservation**
15. Gene sanctuaries are the type of..... **in-situ conservation**
16. The natural protected area for the conservation of genetic diversity is called..... **gene sanctuary**
17. Area of great genetic diversity are protected from human interface are known as **gene sanctuary**
18. Which of the following statement truly describe the biosphere reserve..... **All given** (Biosphere reserves are areas of terrestrial and coastal ecosystems promoting solutions to reconcile the conservation of biodiversity with its sustainable use)

TOPIC NO#8

19. _____can be used to introduce species that left an area..... **Ex-situ conversion**

TOPIC NO#9

20. Plant diversity is urgently & effectively conserved is an objective of..... **GSPC**

TOPIC NO#12

21. _____is a pioneer species, easily regenerated from seed..... **Acacia nilotica**

TOPIC NO# 13-14

22. Pakistan in kinow production..... **Six largest producer**

TOPIC NO#15

23. Peste de petits ruminants (PPR) also known as..... **sheep & goat plague**

TOPIC NO#16-17

24. Sheep was domesticated by humans in..... **10,000BC**

TOPIC NO#18

25. Snow leopard population in Pakistan left **200**
26. Himalayan brown bear population in Pakistan..... **150-200**

TOPIC NO#19

27. Seaweeds population is..... **13000**
28. Which disease is spread due to consuming mercury-poisoned fishes by people **MNAMATA disease.**

TOPIC NO# 20-21

29. Closely related individuals..... **inbreeding**
 30. National strategy for PGRFA may help..... **Country**
 31. Some crustaceans migrate for..... **breeding**

TOPIC NO# 22

32. NCCP stands for **national culture collection of Pakistan**
 33. Microorganisms are..... **more diverse**
 34. ____ is defined as the number of species represented in a specific region landscape or an ecological community..... **Species richness**

TOPIC NO# 24

35. Anthropogenic impact is apparent in the **Coastal zone**
 36. Woody plants like trees have a -- genetic diversity than vascular plants like grasses..... **Higher**

TOPIC NO# 28-30

37. GEWIS stand for..... **A genome wide interaction scan**
 38. A _____ can be taken to analyses G across E interactions..... **polygenic approach**
 39. Which of the following approach can be taken to analyze GxE interaction..... **polygenic**
 40. Phenylketonuria caused by..... **Mutation**

TOPIC NO#33

41. The strongest effect of genetic drift inn..... **small population**

TOPIC NO#34

42. More genetic variation in populations..... **natural selection**
 43. If the advantage gene is removed then the effect is called..... **genetic drift**

TOPIC NO#35-38

44. Bottleneck effect elephant seal..... **Over hunting**

TOPIC NO#40

45. Mild breeding is type of..... **inbreeding**

TOPIC NO#42

46. Triggers of migration..... **all (. local climate 2. local availability of food 3. the season of the year 4. for mating reasons)**

TOPIC NO#43

47. Crow follow migration..... **Daily**

TOPIC NO#45

48. Longest migration was observed in..... **artic tern bird**

49. Wildebeest are the part of **great migration**

TOPIC NO#46

50. There are _____ types of genome transfer **three**

51. _____ can be transferred sexually or asexually **vertical gene transfer**

TOPIC NO#47-48

52. Darwin finches classical example of..... **Adaptive radiation**

53. _____ is an example allopatric speciation..... **Darwin finches**

54. _____ often result of reproductive isolation..... **Sympatric speciation**

TOPIC NO#51

55. Coral reefs are also threatened Reefs are home to ____ of marine animal..... **25%**

TOPIC NO#52

56. _____ produce by dart poison frog..... **Toxins**

TOPIC NO#55-56

57. Sterculia khasiana was endemic tree of **Khasi Hills in Meghalaya in India**

58. Calypha belongs to family _____ spurge family **euphorbiaceae / spurge family**

TOPIC NO#57-58

59. Species that are threatened are sometimes characterized (population dynamics measure of critical dispensation, a mathematical measure of biomass)related to population growth rate..... **all are given**

60. Houghton's goldenrod typically grows on moist (sandy beaches and shallow depressions between low sand ridges along the shoreline)..... **all are given**

TOPIC NO#59

61. Cryopreservation in liquid nitrogen **-196degrees**

62. Cryopreservation is storage of materials at _____ temperature..... **low**

TOPIC NO#61

63. _____ is a strategy wherein samples of animal genetic materials are preserved cryogenically..... **Cry-conservation of animal genetic resources**
64. FAO stands for..... **Food and agriculture organization**
65. What is Dimethyl sulfoxide used for..... **chelating agent**
66. How long a seed remains viable is known as it is **Viability**

TOPIC NO#63

67. OHSS stand for **Ovarian hyper stimulation syndrome**

TOPIC NO#64

68. The 1st report on fish sperm cryopreservation was published by..... **Blaxter (1953).**

TOPIC NO#65

69. Which of the following is the earliest modern protected area?
..... **Yellow stones national park**

TOPIC NO#66

70. The problem in strict nature reserves is (a disease b, climate change, air pollution)..... **Both B & C**

TOPIC NO#68-69

71. WCPA stands for..... **world commission on protected areas**
72. Natural geological and geomorphological features..... **Waterfalls, cliffs, craters, caves etc.**
73. They are generally quite small-protected areas and often have high visitor value category VIII..... **National Monument-Feature**
74. _____ are small sites they focus on more prominent natural features..... **National Monument-Feature**

TOPIC NO#70-71

75. Which category area are not associated with presence and intervention
..... **category-IV**
76. Which category areas are not necessary associated with human presence and intervention..... **Category IV**
77. Main objective of habitat species management area to maintain, conserve and restore species and habitats..... **All of these**
78. Habitat management work..... **all of the above**

TOPIC NO#72-73

79. ____protected areas act as bufferCategory V
80. Which of the following category encourage the conservation of agrobiodiversity and aquatic biodiversity..... Category V
81. The main objective of category V is..... maintain a balance interaction of nature and culture

TOPIC NO#74-75

82. Large natural areas comes into.....category VI
83. The category VI is not designed to accommodate..... large-scale industrial harvest.

TOPIC NO#76-77

84. Largest national park of Pakistan.....Hingol National Park
85. Hingol national park declared reserved in1988
86. Blackbuck became virtually extinct in the Cholistan Desert but the species has been reintroduced in Lal Suhanra
87. When species has been reintroduced in Lal suhanra national park..... Blackbuck

TOPIC NO#78-79

88. Sir Sadiq Muhammad Khan established.....Bahawalpur Zoo
89. Bahawalpur zoo covers.....25 acres

TOPIC NO#82-84

90. In 2013,____rasmar sites has been declared in Pakistan.....19
91. Ramsar conversion was negotiated in ____by countries and NGOs 1960

TOPIC NO#85-86

92. Since the 1970s, the population of the Indus Dolphin has significantly increased here.....1970
93. A sanctuary is a protected area which is reserved for the conservation of only animal and human activities like..... harvesting of timber
94. Chashma and Tuansa barrage declared wild life sanctuaries by..... Punjab government
95. Desert wolf is _____ in Cholistan Wildlife Sanctuary..... Rare
96. Rann of Kutch wildlife sanctuary is part of the..... the desert
97. Sarus crane is one of _____ Species at Runn of Kutch Wildlife sanctuary.....threatened

98. Threatened species of Rann of Kutch is..... **All (Great Indian bustard, Houbara bustard, Sarus crane)**

99. Which of the following statement is correct about the sanctuary.....
A reserved area meant for preservation and development of endangered species

TOPIC NO#87

100. If hunting is prohibited, a game reserve may be considered a” **Nature Reserve”**

101. _____an area wherein controlled hunting and shooting is permitted on permit basis” is known as.....**game reserve**

102. In the_____the, major focus is specifically the animals..... **game reserves**

103. In the game reserves, the major focus is specifically the..... **animals.**

TOPIC NO#88

104. The Buler’s Fruit bat is listed as _____ on the IUCN red list.....**Critically Endangered**

105. The IUCN has listed Great Indian Bustard as_____in 2011.....
Critically Endangered

106. Madagascar Pochard die at young age due to..... **All of them (absence of adequate food items and too deep water)**

TOPIC NO#89

107. IBPGR stand for..... **International Board for Plant Genetic Resource**

108. In _____conservation of habitats rich in genetic diversity was recommended in the UN conference..... **1972**

109. an International Board for Plant Genetic Resource (IBPGR) was established in..... **1972**

110. Plant seeds are stored in _____for long-term storage..... **In-vivo Gene Bank**

111. Germplasm conservation is important to conserve the _____ plant..... **Wild**

TOPIC NO#91

112. High humidity increases _____growth..... **fungal**

113. Vegetative propagation is..... **tissue culture**

114. Tissue culture conservation is an alternative to..... **Vegetative propagation.**

115. The removal of the medium and transfer of cells from its previous culture into fresh growth medium for further propagation of the cell is called.....

Sub culturing

116. During tissue culture growth room temperature range from_____depending on species requirements..... **22°C to 28°C**

117. Suitable light intensity range must of the plant culture is..... **10 to 1000 $\mu\text{mol S}^{-1} \text{m}^{-2}$.**

118. Light intensity for most plants culture require **50-200 $\mu\text{mol-S}^{-1}\text{-m}^{-2}$**

TOPIC NO#92

119. The useful of the specimen for PCR based assays is greater than 4 years when stored at..... **-80 C°**

120. DNA Banks are used for the storage of those species that cannot be conserved in process of conservation..... **in situ**

121. DNA of plant can be preserved by rapid drying of plant samples with the help of _____material..... **silica gel**

122. Lyophilisation during DNA storage process helps to _____ the DNA..... **preserve**

123. _____is the secure long-term storage of an individual's genetic material..... **DNA bank**

124. _____can now be considered as a means of complimentary conservation..... **DNA banks**

125. The first genomic libraries were cloned in **Plasmid**

TOPIC NO#93-95

126. Gene sanctuaries provide way to preserve (a. wild species, natural ecosystem. Natural habitat) **all of these**

127. Ex-situ conservation is also known as..... **static conservation**

128. In _____gene sanctuaries for conservation of wild relatives of coffee was setup in 1984..... **Ethiopia**

129. Gene sanctuaries not only preserve the existing genetic diversity present in a population due to which new _____ combination would appear with time..... **Allele and gene**

130. Gene sanctuary is generally established in the centres of **diversity**

131. There are two main drawbacks of gene sanctuary one of which is that..... **It cannot be easily maintained**

TOPIC NO#96

132. ____ is a relatively cheap method for conserving a broad range of germplasm..... **Seed storage**

TOPIC NO#97

133. Invertebrates accounts for ____% of the animals on earth..... **95%**

134. ____ Organisms are described as engineers of ecosystem..... **earthworms**

135. It has been at least ____ of world food production comes from crops that are dependent on insect pollination..... **35 percent**

136. At least 35 percent of world food production comes from crops that are dependent on..... **Insect pollination.**

137. Some larger soil-dwelling invertebrates, such as earthworms, ants and termites help maintain the chemical fertility needed for **plant growth**

138. Some larger soil-dwelling invertebrates, such as earthworms, ants and termites, have been described as **“ecosystem engineers”.**

139. Perhaps the most neglected group of all – in research, in farming practices, and in policies and strategies for agriculture and biodiversity – are the soil-dwelling **invertebrates**

TOPIC NO#98

140. _____ have created barriers to the migration of invertebrate species..... **Human activities**

141. ____ is Key factor for regulating many of bio chemical process **temperature**

142. Most invertebrates are expected to change their geographical distribution..... **in response to climate change**

TOPIC NO#99

143. ____ help to maintain soil structure and the availability of water throughout the soil profile..... **Earthworms**

144. This lack of diversity may inhibit the ability of the population to respond to **climate change.**

145. The capacity of ecosystems to adapt is diminished when biodiversity of ____ is lost..... **invertebrate**

146. The adaptive capacity of ecosystem is more likely to dependent on..... **invertebrate biodiversity**

147. Studies have shown that the presence of these animals can help to.....
alleviate

TOPIC NO#100

148. _____are needed to be able to cope with changing climatic conditions, and increasing demands for food and retain the capacity to adapt to potential changes in the types of crops grown.....Pollinator populations

149. Considering the better studied mammals and birds 100% of the currently described species have been evaluated for their conservation statuses and out of these _____ out of 9,990 birds species are considered to be endangered.....12%

150. Endemic species are..... Species localized in a specific region

TOPIC NO#101

151. Though forest trees are known for showing great _____in their response to climate changes..... plasticity

152. Forest genetic resources or tree genetic resources are _____of shrub and tree species of actual or future value..... genetic material

153. In _____the Food and Agriculture Organization of the United Nations published the first State of the World's Forest Genetic Resources2014

154. Diversity of forest genetic resources enables the species to adapt to..... All given (temperature changes, drought, pests, diseases and forest fires)

155. Forest help in driving.....Climate Change

156. Forests are important due to.....all given (stand, population or landscape of trees, and typically other associated woody plants)

TOPIC NO#102

157. Forests are important due to..... (All given)

158. Changes in climate also make it possible for _____and diseases to invade new areas, destroying the forests there..... Pests

159. Forests majorly remove _____from the atmosphere and produce oxygen..... Carbon dioxide

160. In Europe one estimate put the role of value marketed _____ non wood goods from forest at..... €2.3 billion

161. Globally, forests remove _____from the atmosphere and produce oxygen..... carbon dioxide

TOPIC NO#103

162. Climate has direct effect on **FGR**
163. Tree mortality in ecosystem increase due to change in..... **all given (extreme climatic events, particularly drought in combination with widespread regeneration failure)**

TOPIC NO#104

164. The ability of one genotype to produce more than one phenotypes when exposed to different environment conditions without genetic change is called..... **phenotypic plasticity**
165. _____ is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions..... **Phenotypic plasticity**
166. Tree populations rely on three interplaying mechanisms to respond to environmental change..... **(All given) adaptation, Migration, phenotypic plasticity**

TOPIC NO#105

167. Hybridisation may also introduce the new genetic variation required by trees to adapt to..... **Novel environments.**
168. Some insects use tree terpenes as precursors for their communication_____ or incorporate them into their own defense systems..... **pheromones**
169. Natural selection favors genotypes with **(All given) increased tolerance or resistance to disturbances, and phenotypic plasticity**
170. ____describes a situation where two or more) species reciprocally affect each other's evolution..... **Co-evolution**

TOPIC NO#106

171. Climate change-related traits including..... **(All given) plasticity and adaptation**
172. Candidate genes for drought tolerance include those involved in..... **(All given) the synthesis of abscisic acid, transcriptional regulators**
173. Using microarray analysis, detected that up to _____were significantly induced by drought in two Mediterranean pine species..... **113 genes**
174. To date, one of the most extensive programmers to develop trees with resistance to insect pests in temperate regions is in..... **British Columbia.**

175. The act of moving plants by human to different habitat to experience different environment conditions is term as..... **Assisted migration**

176. Assisted migration may be undertaken.....**all given (over long distances, current range limit of particular genotypes and populations, or within the existing range.**

TOPIC NO#107

177. _____ Seed orchards, clone banks and clonal archives are examples of _____conservation units..... **static ex situ**

178. _____ensures the success of species in environments that are highly variable and subject to change..... **Genetic diversity**

TOPIC NO#108-109

179. _____will be fine for cutting tree in 1992 act **5000 rupees**

180. The Wild Birds and Animals Protection Act..... **1912**

TOPIC NO#110

181. Convention on biological diversity..... **1992**

TOPIC NO#113

182. Cartagena protocol was adopted on **June 2001 in Cartagena, Spain.**

183. Cartagena protocols not cover products derived from..... **LMOs**

184. Cartagena protocols deals with..... **biosafety**

TOPIC NO#117

185. The Pakistan Animal Quarantine Ordinance..... **1979**

TOPIC NO#118

186. (IABGR) stand for **Institute of Agricultural Biotechnology and Genetic Resources**

187. Number of lab in plant genetic resource in Pakistan..... **six**

TOPIC NO#120

188. Greater efforts are needed to estimate the full value of..... **PGRFA,**

189. In plant tissue culture what is the term ORGANOGENESIS means..... **Formation of root and shoot from callus culture**
190. Plant are totipotent because..... **A single cell could become a complete plant**
191. The type of coiling in DNA is **right-handed**
192. ____ is not generally seen in biodiversity hotspots..... **Lesser interspecific competition**
193. ____ Factor/factors can affect the services of invertebrates to the environment..... **ALL Given**

BEST OF LUCK

LAST FIGHT OF WARRIORS